Shot Boundary Detection

Topic in Computer Vision

Contents

- What is a shot?
- Types of shot break
- Shot boundary detection?
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What is Shot?

- Shot is the result of uninterrupted camera work
- Shot-break is the transition from one shot to the next

Types of Short Break

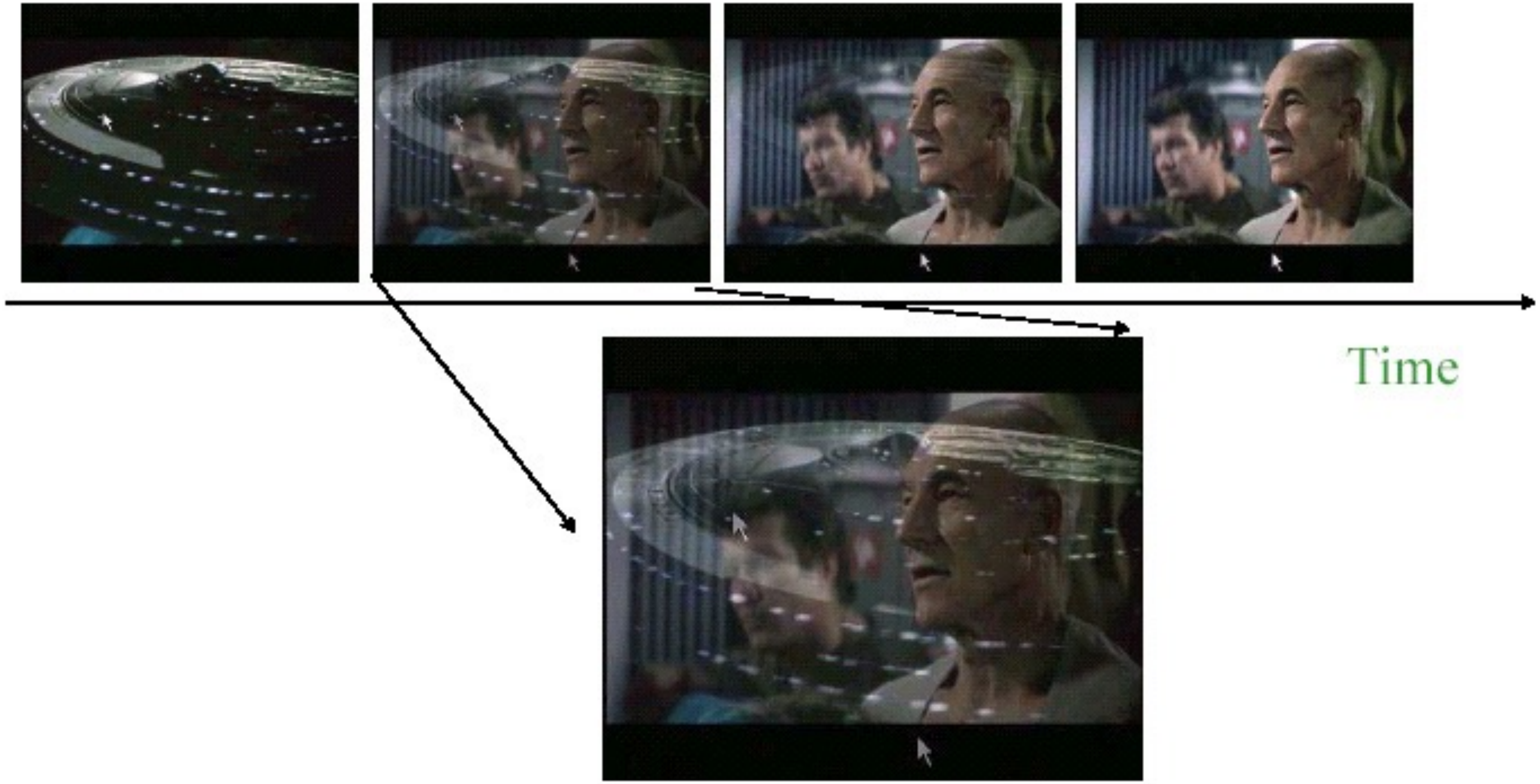
Shot-Break

- Hard Cut
- Fade
- Dissolve
- Wipe

Hard Cut

Fade

Dissolve



Wipe

Shot Detection

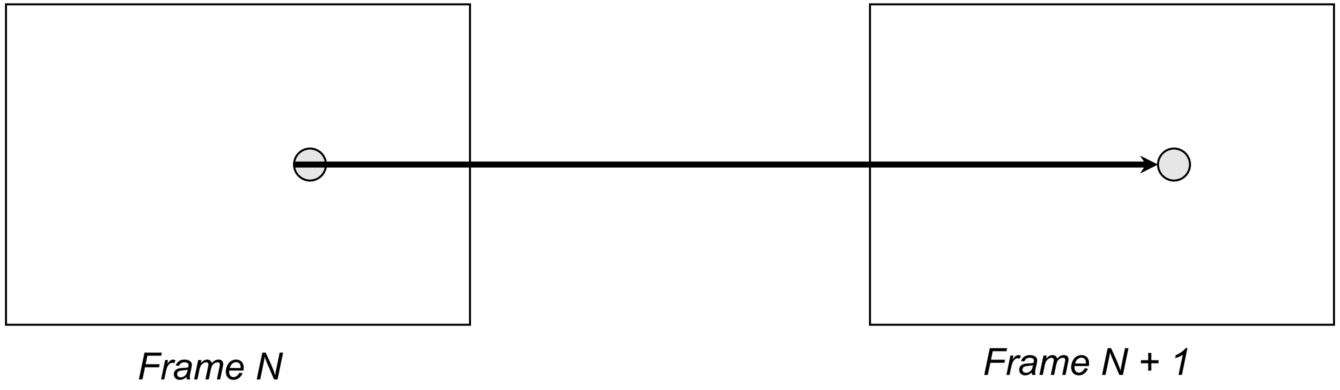
Shot Detection

- Goal
  - To segment video into shots
- Two ways:
  - Cluster the similar frames to identify shots
  - Find the shots that differ and declare it as shot-break

Approaches for Shot Detection

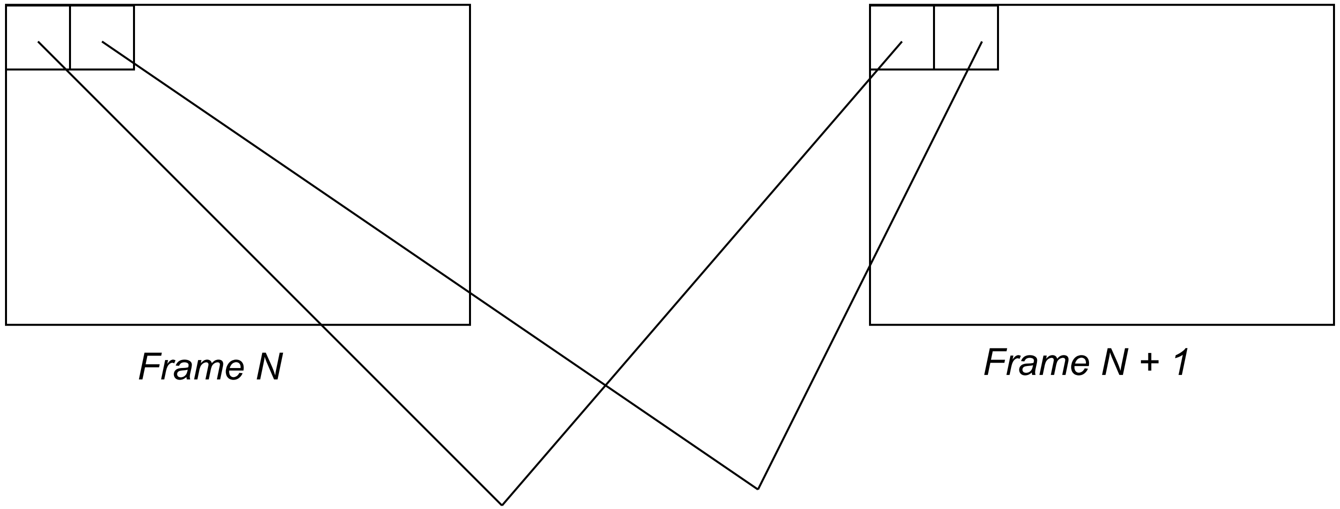
- Pixel Comparison
- Block-based approach
- Histogram Comparison
- Edge Change Ratio

Pixel Comparision



$$D(i,i+1) = \frac{\sum_{x=1}^{X} \sum_{y=1}^{Y} | P_i(x,y) - P_{i+1}(x,y) |}{XY}$$

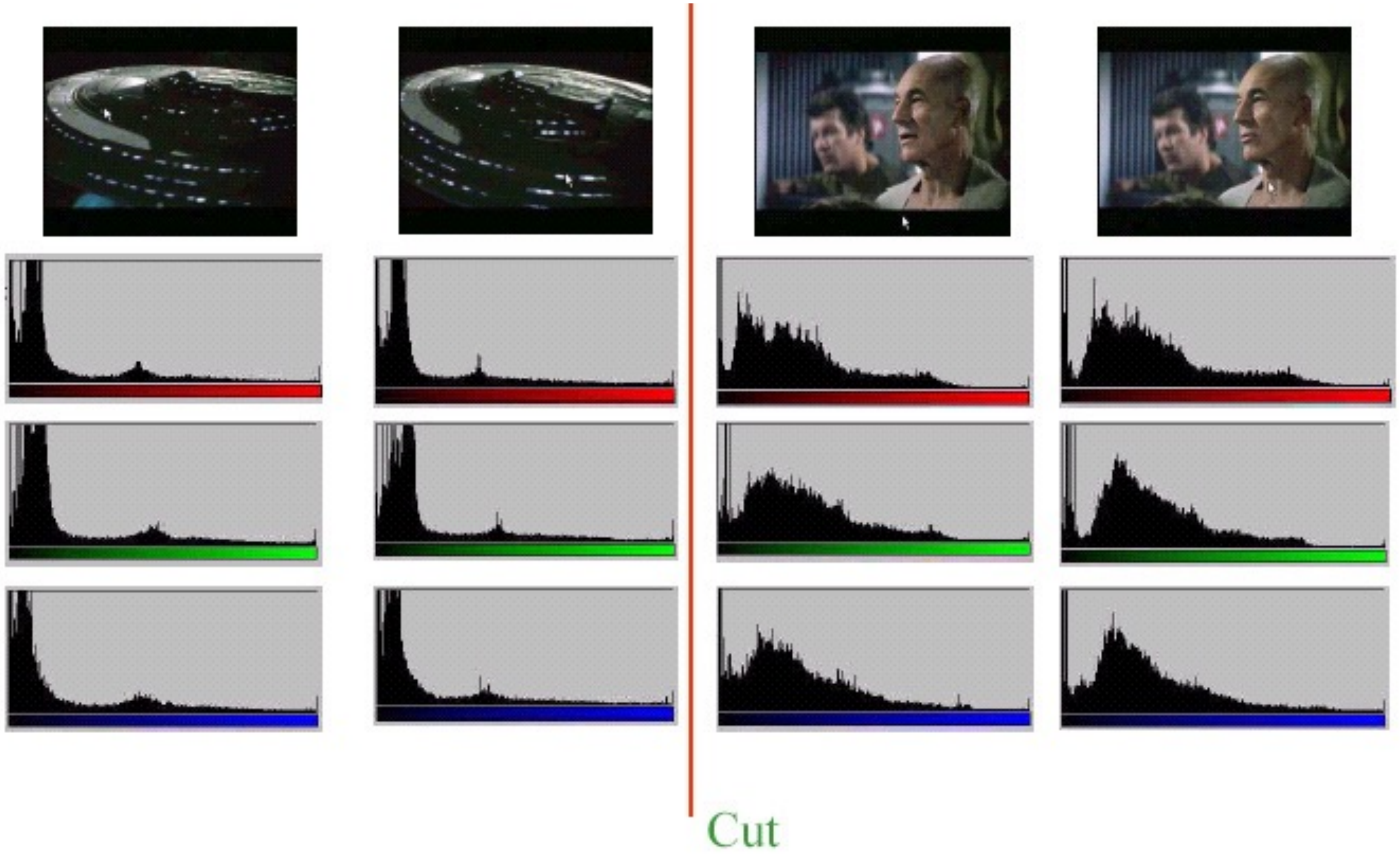
Block Based Approaches



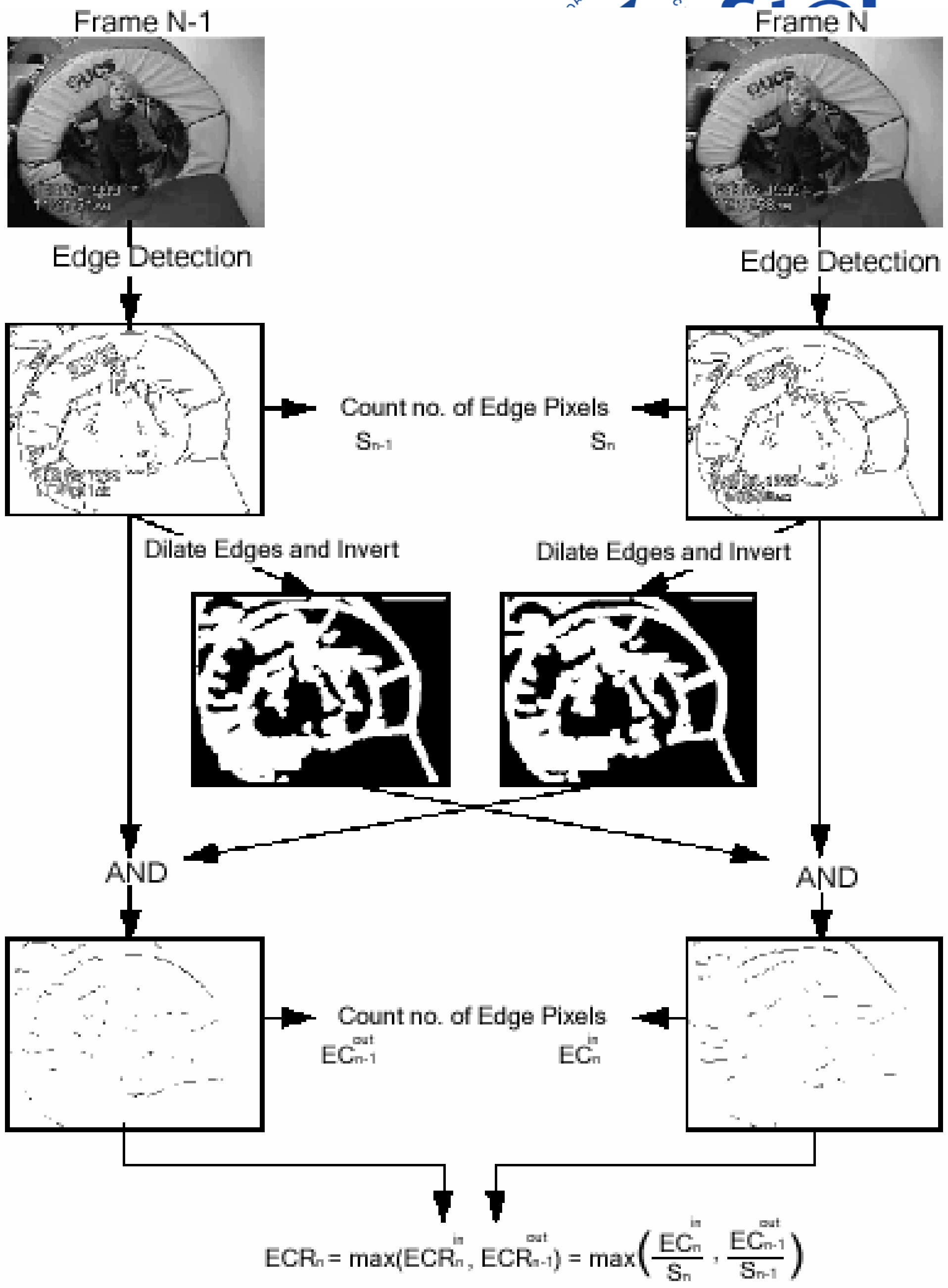
Compares statistics of the corresponding blocks

Counts the number of significantly different blocks

Histogram



Edge Change Ratio



Comparison

Method	Advantages	Disadvantages
Pixel-Comparison	Simple, easy to implement	Computationally heavy, Very sensitive to moving object or camera motion
Block based	Performs better than pixel	Can't identify dissolve, fade, fast moving objects
Histogram comparison	Performance is better Detects hard-cut, fade, wipe and dissolve	Fails if the two successive shots have same histogram. Can't distinguish fast object or camera motion
Edge Change Ratios	Detects hard-cut, fade, wipe and dissolve	Computationally heavy Fails when there is large amount of motion

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Applications

- Video Summarization
- Video Streaming

Questions and Answers